

Four-Corner Spectral Tuning for Search-Free ADMM in Diffeomorphic Image Registration

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Abstract

The choice of penalty and relaxation parameters strongly affects the runtime of ADMM-based variational image registration. For a well-posed quadratic subproblem every admissible pair is already contractive, so the practical task is to select a rapidly convergent pair without a numerical spectral search. We show that the rank-one pixelwise registration Hessian and the scalar Sobolev symbol reduce constant-model tuning exactly to four curvature–symbol corners, which yields a finite algebraic one-shot predictor. On all 134 FIRE retinal pairs at 256×256 , the predictor reduced median pairwise runtime by 34.4% and inner iterations by 39.6% relative to an externally validation-selected fixed pair, while landmark error and topology were unchanged; it was also 51.7% faster than a safeguarded Barzilai–Borwein proxy. On ten CIMA histology pairs, matched-protocol runtime decreased by 20.1%. Signed variable-coefficient enclosures supply optional rigorous rate bounds that are not used during tuning.

Keywords: alternating direction method of multipliers, diffeomorphic image registration, parameter selection, spectral analysis, local Fourier analysis

1 Introduction

Diffeomorphic image registration estimates an invertible, topology-preserving spatial transformation that aligns a moving image with a fixed image. Classical variational methods remain attractive because they provide explicit control over the deformation model, the regularization, and the numerical stopping criteria, and because they can be applied without training data. Representative approaches include DARTEL, diffeomorphic demons, large-deformation optimal-control formulations, and modern GPU-accelerated variational solvers [1, 10, 14, 15]. Even as learning-based registration has matured [7], the main practical limitation of these classical solvers remains computational cost.

Many diffeomorphic schemes update a velocity field through a sequence of convex, linearized optical-flow problems. A typical quadratic subproblem is

$$\min_{v=(v_x, v_y)} \frac{\mu}{2} \|I_x \odot v_x + I_y \odot v_y + I_t\|_2^2 + \frac{1}{2} \|Lv\|_2^2, \quad (1)$$

where $\mu > 0$ is a data-fidelity weight, I_x and I_y are spatial image derivatives, I_t is the current temporal or residual derivative, $\odot$ denotes the pointwise product, and L is a differential regularization operator. Splitting the data and regularization terms produces inexpensive pixelwise and Fourier-domain subproblems, which makes ADMM particularly suitable for GPU implementation [12, 14].

In this splitting, the ADMM penalty $\rho > 0$ scales the quadratic constraint term that couples the data and regularization subproblems, while the over-relaxation parameter $\alpha \in (0, 2]$ mixes the most recent primal update into the dual and secondary steps; both strongly affect the iteration’s contraction rate and therefore the practical runtime. Fixed defaults, residual balancing, and adaptive spectral rules can all be effective, but none of them exploits the unusually specific geometry of registration: the data Hessian is block diagonal with rank-one 2×2 blocks, whereas the regularizer is a scalar elliptic operator applied identically to each velocity component. General quadratic ADMM theory yields optimal parameters for several special classes [5, 6, 13]. More recently, Song et al. optimized ρ numerically for general linear quadratic problems and derived an optimal α conditional on ρ , including experiments on diffeomorphic registration [12]; their registration penalty still requires a spectral optimization.

Our approach is based on local Fourier analysis (LFA), a classical tool for predicting and optimizing stationary iterations for constant-coefficient differential operators [4, 16]. Standard LFA is exact for translation-invariant periodic problems, and nonstandard variants remain predictive for jumping and random coefficients [9, 11]. The rank-one image-gradient blocks admit a sharper global reduction: the constant-model tuning envelope depends only on the two attained data-curvature endpoints and the two attained regularizer-symbol endpoints. This yields four branches, a finite algebraic candidate set, and no numerical search over the penalty.

Contributions.

1. An exact reduced Cayley representation and a strict-contraction characterization for well-posed quadratic registration subproblems (Section 4).
2. An exact four-corner reduction and a finite algebraic, search-free joint predictor for rank-one registration Hessians (Sections 5 and 7).
3. Optional signed variable-coefficient rate enclosures for nonnormal iterations (Section 6).
4. FIRE and CIMA validation demonstrating lower end-to-end runtime at matched landmark accuracy and topology (Section 9).

Organization.

Section 2 reviews related work. Section 3 fixes the discrete registration model and the ADMM splitting, and Section 4 derives the exact reduced iteration. Section 5 establishes the four-corner reduction, Section 6 develops the optional rate certificates, and Section 7 gives the finite parameter selector. Sections 8 and 9 describe the implementation and the experiments; Sections 10 and 11 discuss the findings and conclude.

2 Related Work

2.1 ADMM parameter selection

The scaled ADMM iteration and its practical residual stopping criteria were systematized by Boyd et al. [2]. Ghadimi et al. derived optimal penalties and relaxation parameters for several quadratic classes [5], and Teixeira et al. studied scaling, preconditioning, and optimal ADMM parameters in distributed quadratic programming [13]. Giselsson and Boyd developed tight contraction bounds and metric selection for Douglas–Rachford splitting and ADMM under strong-convexity and smoothness assumptions [6]. Adaptive alternatives include residual balancing and Barzilai–Borwein (BB) type spectral updates [17, 18].

The closest work is that of Song et al. [12], who formulate ADMM and over-relaxed ADMM (oADMM) for general linear quadratic problems, optimize the penalty from the iteration spectrum, give a closed-form relaxation parameter for a fixed penalty, and include diffeomorphic image registration as an application. Our objective differs in kind. Instead of a full-spectrum criterion for a general quadratic program, we exploit the specific geometry of a rank-one registration data term paired with a scalar Sobolev regularizer. That geometry replaces their numerical one-variable spectral optimization of the penalty by a finite four-corner predictor, and their conditional relaxation formula by a finite set of piecewise-linear candidates; neither stage requires gradient descent, repeated ADMM trials, or the full iteration spectrum. Spatial coefficient variation is likewise treated differently: rather than forming the exact spectrum of the full iteration, we combine a local predictor with a separately evaluated signed enclosure that assumes no commutativity. The resulting selector

is assessed empirically by matched-accuracy run-time reductions on the FIRE and CIMA landmark benchmarks.

2.2 ADMM and variational diffeomorphic registration

DARTEL [1] and diffeomorphic demons [15] are influential optimization-based registration methods. Thorley et al. proposed a fast diffeomorphic method that composes nonstationary velocity updates and solves arbitrary-order regularized convex subproblems with accelerated ADMM on GPUs [14]. Song et al. later used a closely related quadratic registration model to evaluate spectrally optimized ADMM and oADMM [12].

2.3 Local Fourier analysis

Fourier analysis has long been used to estimate convergence factors of stationary iterations for elliptic problems [4], and LFA became a principal tool for multigrid design and parameter optimization [16]. Its exactness under periodic translation-invariant assumptions, and its extensions to broader settings, have been studied theoretically [11], and nonstandard LFA accurately predicts multigrid behavior for random and discontinuous coefficients [9]. Our use of LFA differs in two respects: the iteration is an ADMM splitting map rather than a smoother, and the variable coefficient is a matrix-valued, rank-one optical-flow Hessian.

3 Registration Model and ADMM Splitting

3.1 Discrete quadratic model

Let the image domain contain n grid points in dimension $d \in \{2, 3\}$, let Δx denote the grid spacing, and stack the d velocity components into $v \in \mathbb{R}^{dn}$. Write I for the linearized image intensity and, at grid point x_i , let $q_i = \nabla I(x_i) \in \mathbb{R}^d$ be its spatial gradient. With data weight $\mu > 0$, and allowing an optional isotropic damping $\nu \geq 0$ already present in some implementations, the linearized data Hessian is the block-diagonal matrix

$$\begin{aligned} H &= \text{diag}(H_1, \dots, H_n), \\ H_i &= \nu I_d + \mu q_i q_i^\top \succeq 0. \end{aligned} \quad (2)$$

When $\nu = 0$ this recovers the pure rank-one optical-flow Hessian. Each $d \times d$ block has eigenvalues ν (with multiplicity $d - 1$) and $\nu + \mu \|q_i\|^2$. Throughout, Id denotes the identity of conforming size on $\mathbb{R}^{dn}$ unless a subscript such as I_d or I_n indicates otherwise.

We assume a componentwise Sobolev regularizer. Let $G_0 \in \mathbb{R}^{n \times n}$ be a symmetric positive semidefinite discretization of a scalar elliptic operator on the n -point grid (finite-difference or spectral), and let I_d denote the $d \times d$ identity. The full regularizer on the stacked velocity is then the Kronecker product

$$G = L^\top L = G_0 \otimes I_d \in \mathbb{R}^{dn \times dn}, \quad (3)$$

which applies the same spatial operator G_0 independently to each of the d velocity components; equivalently, L is a differential regularization operator acting componentwise. A small zeroth-order term γI_n , with weight $\gamma \geq 0$ and I_n the $n \times n$ identity, or an appropriate boundary condition, may be included in G_0 to remove its common nullspace.

When G_0 is translation-invariant and periodic, it is diagonalized by the discrete Fourier transform. Its eigenvalues are then given by a scalar Fourier symbol $g(\xi)$, indexed by discrete frequencies $\xi \in \mathbb{R}^d$. For a first-order Sobolev regularizer with stiffness $\beta > 0$,

$$g(\xi) = \gamma + \frac{4\beta}{(\Delta x)^2} \sum_{\ell=1}^d \sin^2\left(\frac{\xi_\ell}{2}\right). \quad (4)$$

Thus $\text{spec}(G_0) = \{g(\xi)\}$ over the discrete frequencies, and $\text{spec}(G) = \text{spec}(G_0)$ with multiplicity d . In particular, the spectral endpoints used later satisfy $g_- = \lambda_{\min}(G_0) = \min_\xi g(\xi)$ and $g_+ = \lambda_{\max}(G_0) = \max_\xi g(\xi)$.

The registration subproblem is

$$\min_v \frac{1}{2} v^\top H v + b^\top v + \frac{1}{2} v^\top G v, \quad (5)$$

where $b \in \mathbb{R}^{dn}$ stacks the linearized optical-flow residuals (in two dimensions, the i th block is proportional to $\mu I_t(x_i) q_i$, up to sign convention). We

split (5) by introducing an auxiliary copy $w = v$:

$$\begin{aligned} \min_{v,w} f_H(v) + f_G(w) \quad \text{subject to} \quad v - w = 0, \\ f_H(v) := \frac{1}{2}v^\top H v + b^\top v, \quad f_G(w) := \frac{1}{2}w^\top G w. \end{aligned} \quad (6)$$

3.2 ADMM and over-relaxation

Let $u \in \mathbb{R}^{dn}$ denote the scaled dual variable associated with the constraint $v - w = 0$. Standard ADMM applied to (6) is

$$(H + \rho \text{Id})v^{k+1} = -b + \rho(w^k - u^k), \quad (7)$$

$$(G + \rho \text{Id})w^{k+1} = \rho(v^{k+1} + u^k), \quad (8)$$

$$u^{k+1} = u^k + v^{k+1} - w^{k+1}. \quad (9)$$

For oADMM, v^{k+1} is replaced in the w and dual updates by

$$\hat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha)w^k, \quad 0 < \alpha \leq 2. \quad (10)$$

The linear term b shifts the fixed point but does not affect error propagation, so all spectral statements below concern the homogeneous iteration. We write $T_{\rho,\alpha}$ for the resulting linear error map on the state (w, u) , defined by the homogeneous form of (7)–(9) with the relaxation (10).

4 Exact Reduced Iteration

Define the resolvents and Cayley transforms

$$P_H(\rho) = \rho(H + \rho \text{Id})^{-1}, \quad (11)$$

$$C_H(\rho) = (H - \rho \text{Id})(H + \rho \text{Id})^{-1} = \text{Id} - 2P_H(\rho), \quad (12)$$

$$P_G(\rho) = \rho(G + \rho \text{Id})^{-1}, \quad (13)$$

$$C_G(\rho) = (G - \rho \text{Id})(G + \rho \text{Id})^{-1} = \text{Id} - 2P_G(\rho). \quad (14)$$

Proposition 4.1 (Reduced Cayley iteration). *For the quadratic splitting (6), all nonzero eigenvalues of the standard ADMM error matrix $T_{\rho,1}$ coincide with the eigenvalues of*

$$E_{\rho,1} = \frac{1}{2}(\text{Id} + C_H(\rho)C_G(\rho)). \quad (15)$$

For oADMM the corresponding reduced operator is

$$E_{\rho,\alpha} = \left(1 - \frac{\alpha}{2}\right) \text{Id} + \frac{\alpha}{2} C_H(\rho)C_G(\rho), \quad (16)$$

and consequently $\varrho(T_{\rho,\alpha}) = \varrho(E_{\rho,\alpha})$, where ϱ denotes the spectral radius.

The proof is given in Appendix A. The product $C_H C_G$ is generally nonnormal because H and G do not commute, which is precisely why a global simultaneous diagonalization is unavailable for registration.

Theorem 4.2 (Contraction is independent of tuning). *Let $H, G \succeq 0$ and $H + G \succ 0$. For every $\rho > 0$,*

$$\kappa_\rho := \|C_H(\rho)C_G(\rho)\|_2 < 1. \quad (17)$$

Consequently, for every $0 < \alpha \leq 2$,

$$\varrho(E_{\rho,\alpha}) \leq \|E_{\rho,\alpha}\|_2 \leq 1 - \frac{\alpha}{2}(1 - \kappa_\rho) < 1. \quad (18)$$

Proof For a symmetric positive semidefinite matrix A , the Cayley eigenvalue at $\lambda \geq 0$ is $(\lambda - \rho)/(\lambda + \rho)$, whose modulus is at most one, with equality only at $\lambda = 0$. Thus $\|C_A x\|_2 \leq \|x\|_2$, with equality exactly when $x \in \ker A$. If equality held for $C_H C_G$ on a unit vector x , then equality in both inequalities $\|C_H C_G x\|_2 \leq \|C_G x\|_2 \leq \|x\|_2$ would give $x \in \ker G$, hence $C_G x = -x$ and $C_G x \in \ker H$. Therefore $x \in \ker H \cap \ker G$, contradicting $H + G \succ 0$. Compactness of the unit sphere gives $\kappa_\rho < 1$. The oADMM bound follows from the triangle inequality applied to (16), whose coefficients are nonnegative for $0 < \alpha \leq 2$. $\square$

Nullspaces and conventions.

If a nonzero x lies in both kernels, then $C_H x = C_G x = -x$, $C_H C_G x = x$, and $E_{\rho,\alpha} x = x$. This is a direction along which the objective is nonunique, not a tuning-induced instability. Registration removes it by zeroth-order regularization, by appropriate boundary conditions, or by restriction to the identifiable subspace. The endpoint $\alpha = 2$ is admissible for the reduced linear contraction above; implementations whose standard oADMM theorem assumes $\alpha < 2$ should retain that conventional restriction.

5 Four-Corner Spectral Tuning

5.1 Patch surrogate

Partition the image grid into R patches $\{\Omega_r\}_{r=1}^R$. On patch r , choose a representative image-gradient Hessian

$$H_r = \mu \bar{q}_r \bar{q}_r^\top + \nu I_d, \quad (19)$$

where $\bar{q}_r$ may be a center, mean, or robust medoid gradient and $\nu \geq 0$ is the optional damping from (2). Denote its eigenvalues by $h_{r,1} \leq \dots \leq h_{r,d}$; for the rank-one model, $h_{r,1} = \dots = h_{r,d-1} = \nu$ and $h_{r,d} = \nu + \mu \|\bar{q}_r\|^2$.

Let G_r be a translation-invariant patch regularizer, implemented with periodic patch boundary conditions or another transform-diagonalizable closure, and write $g_r(\xi)$ for its scalar Fourier symbol. Define the attained symbol endpoints

$$g_r^- := \min_{\xi} g_r(\xi), \quad g_r^+ := \max_{\xi} g_r(\xi), \quad (20)$$

so that $g_r(\xi) \in [g_r^-, g_r^+]$. The block-direct-sum surrogate is

$$\tilde{H} = \bigoplus_r (I_{|\Omega_r|} \otimes H_r), \quad \tilde{G} = \bigoplus_r (G_r \otimes I_d). \quad (21)$$

We write $\tilde{E}_{\rho,\alpha}$ for the reduced oADMM operator (16) formed with $(\tilde{H}, \tilde{G})$ in place of (H, G) .

5.2 Exact patch symbol

Because H_r acts only on velocity components while G_r has a scalar spatial symbol, the two commute on a patch. This gives the frozen-patch spectrum in closed form.

Theorem 5.1 (Frozen-patch spectrum). *The reduced oADMM surrogate $\tilde{E}_{\rho,\alpha}$ is normal, and its eigenvalues are*

$$e_{r,j}(\xi; \rho, \alpha) = 1 - \frac{\alpha}{2} + \frac{\alpha}{2} \frac{h_{r,j} - \rho}{h_{r,j} + \rho} \frac{g_r(\xi) - \rho}{g_r(\xi) + \rho}. \quad (22)$$

Equivalently, writing

$$\theta(h, g; \rho) = \frac{\rho^2 + hg}{(\rho + h)(\rho + g)}, \quad (23)$$

we have $e_{r,j}(\xi; \rho, \alpha) = 1 - \alpha + \alpha \theta(h_{r,j}, g_r(\xi); \rho)$, so that the exact frozen-patch envelope is

$$U_{\text{LFA}}(\rho, \alpha) = \max_{r,j,\xi} |e_{r,j}(\xi; \rho, \alpha)|. \quad (24)$$

5.3 Corner reduction

The continuous maximization over frequency in (24) is unnecessary.

Lemma 5.2 (Corner principle). *For fixed $\rho > 0$ and scalar arguments $h, g \geq 0$,*

$$\begin{aligned} \frac{\partial \theta}{\partial h} &= \frac{\rho(g - \rho)}{(g + \rho)(h + \rho)^2}, \\ \frac{\partial \theta}{\partial g} &= \frac{\rho(h - \rho)}{(h + \rho)(g + \rho)^2}. \end{aligned} \quad (25)$$

Hence θ is coordinatewise monotone on every rectangle $[h^-, h^+] \times [g^-, g^+]$, and its extrema occur at the four corners. Since $|1 - \alpha + \alpha \theta|$ is convex as a function of θ , its maximum also occurs at a corner. In the present setting the second argument ranges over regularizer eigenvalues, i.e. values of the symbol $g(\xi)$ or, equivalently, of $\text{spec}(G_0)$.

Theorem 5.3 (Global four-corner reduction). *Let $H_i = \nu I_d + \mu q_i q_i^\top$ as in (2), let $g_- = \lambda_{\min}(G_0)$ and $g_+ = \lambda_{\max}(G_0)$ be the attained extrema of the discrete scalar regularizer spectrum, and put*

$$h_- = \nu, \quad h_+ = \nu + \mu \max_i \|q_i\|^2. \quad (26)$$

For every $\rho > 0$ and every $0 < \alpha \leq 2$, the envelope over all pixelwise rank-one curvatures and all regularizer frequencies equals

$$\max_{(h,g) \in \{h_-, h_+\} \times \{g_-, g_+\}} |1 - \alpha + \alpha \theta(h, g; \rho)|. \quad (27)$$

Consequently the finite joint constant-model predictor depends on exactly four branches, independently of image size and of the number of distinct gradient magnitudes.

Proof Every data eigenvalue equals either ν or $\nu + \mu \|q_i\|^2$, hence lies in $[h_-, h_+]$, and both endpoints are attained: the lower endpoint corresponds to the direction orthogonal to q_i . By Lemma 5.2, θ maps the rectangle $[h_-, h_+] \times [g_-, g_+]$ onto an interval whose endpoints are attained at its corners. The map $t \mapsto |1 - \alpha + \alpha t|$ is convex, so its maximum over that interval is attained at an endpoint, and therefore at one of the same four corners. The argument permits $h_- = 0$, $g_- = 0$, repeated endpoints, and common nullspaces, because every denominator in (23) is strictly positive for $\rho > 0$. $\square$

More generally, for the patchwise surrogate let

$$\mathcal{C} = \bigcup_{r,j} \{(h_{r,j}, g_r^-), (h_{r,j}, g_r^+)\}. \quad (28)$$

Then

$$U_{\text{LFA}}(\rho, \alpha) = \max_{(h,g) \in \mathcal{C}} |1 - \alpha + \alpha \theta(h, g; \rho)|, \quad (29)$$

so the full local Fourier calculation reduces to a small finite set even in three dimensions. For the global rank-one predictor, Theorem 5.3 sharpens this set to four entries; patchwise sets remain useful only when a local rather than a global model is deliberately constructed.

6 Optional Variable-Coefficient Rate Certification

This section develops rigorous upper bounds on the contraction factor of the full variable-coefficient iteration. The bounds are optional diagnostics: they are evaluated once, after the fact, at the parameters chosen in Section 7, and they play no role in the timed registration algorithm. A reader interested only in the practical method may proceed directly to Section 7.

6.1 Frozen-patch perturbation certificate

We first retain the frozen-patch perturbation certificate as a rigorous but deliberately conservative baseline. It replaces both H and G by their patch surrogates. Define

$$\delta_H = \|H - \tilde{H}\|_2, \quad \delta_G = \|G - \tilde{G}\|_2, \quad (30)$$

$$\delta = \delta_H + \delta_G.$$

For block-diagonal H , the quantity δ_H is the largest patchwise 2-norm of a $d \times d$ block difference and is therefore inexpensive. The regularizer error δ_G can be computed with a few power iterations, or bounded by a row-sum estimate of the patch-boundary stencil discrepancy.

Lemma 6.1 (Cayley Lipschitz bound). *For symmetric positive semidefinite A, B and $\rho > 0$,*

$$\|C_A(\rho) - C_B(\rho)\|_2 \leq \frac{2}{\rho} \|A - B\|_2. \quad (31)$$

Theorem 6.2 (Certified local spectral envelope). *Let U_{LFA} be defined by (29). Then, for $0 < \alpha \leq 2$,*

$$\varrho(T_{\rho, \alpha}) \leq \mathcal{U}(\rho, \alpha) := U_{\text{LFA}}(\rho, \alpha) + \frac{\alpha \delta}{\rho}. \quad (32)$$

No commutativity assumption on the full matrices H and G is required.

The proof is given in Appendix B. The bound separates two effects: U_{LFA} measures the exact contraction of the frozen local models, while $\alpha \delta / \rho$ penalizes spatial coefficient variation and patch-boundary approximation. A larger ρ suppresses the perturbation term, whereas an excessively large ρ worsens the local Cayley balance. This tradeoff makes parameter selection nontrivial and certifiable.

Corollary 6.3 (Guaranteed convergence). *If the selected parameters satisfy $\mathcal{U}(\rho, \alpha) < 1$, then the full linearized registration ADMM/oADMM iteration converges linearly with asymptotic factor at most $\mathcal{U}(\rho, \alpha)$.*

6.2 Signed enclosures with the exact regularizer

We now retain $G = G_0 \otimes I_d$ exactly. Let

$$K_\rho := (G_0 - \rho I_n)(G_0 + \rho I_n)^{-1} \in \mathbb{R}^{n \times n} \quad (33)$$

be the scalar Cayley transform of G_0 , so that $C_G(\rho) = K_\rho \otimes I_d$. Write $C_H = \text{diag}(C_1, \dots, C_n)$, where each $C_i = (H_i - \rho I_d)(H_i + \rho I_d)^{-1}$ is an explicit symmetric $d \times d$ block, so that $(C_H C_G)_{ij} = (K_\rho)_{ij} C_i$. Throughout this subsection set

$$B_{ij}^S = |(K_\rho)_{ij}| \|(C_i + C_j)/2\|_2, \quad (34)$$

$$B_{ij}^Q = |(K_\rho)_{ij}| \|(C_i - C_j)/2\|_2$$

for $i \neq j$, with zero diagonal, and let d_i^- and d_i^+ denote the extreme eigenvalues of the normal diagonal block $(K_\rho)_{ii} C_i$. Collect these scalars into the vectors $d^\pm := (d_1^\pm, \dots, d_n^\pm) \in \mathbb{R}^n$.

Theorem 6.4 (Block-Gershgorin enclosure). *Every eigenvalue of $C_H C_G$ lies in a disk*

$$|z - (K_\rho)_{ii} C_i| \leq \|C_i\|_2 \sum_{j \neq i} |(K_\rho)_{ij}| \quad (35)$$

for some pixel i and some eigenvalue $c_{i,r}$ of C_i . The resulting affine disk envelope

$$U_{\text{BG}}(\rho, \alpha) = \max_{i,r} \left[\left| 1 - \frac{\alpha}{2} + \frac{\alpha}{2} (K_\rho)_{ii} c_{i,r} \right| + \frac{\alpha}{2} \|C_i\|_2 \sum_{j \neq i} |(K_\rho)_{ij}| \right] \quad (36)$$

is a rigorous upper bound on $\varrho(T_{\rho,\alpha})$.

Proof sketch Select the eigenvector block of maximum Euclidean norm. The diagonal block $(K_\rho)_{ii} C_i$ is normal, so its shifted smallest singular value is exactly the distance to its spectrum, and the off-diagonal block row supplies the stated radius. The affine oADMM scaling in (16) transfers the disk enclosure to $T_{\rho,\alpha}$. $\square$

The same block structure yields a signed rectangle, which is sharper because it retains the sign of the real part. Writing $M := C_H C_G$, let $S = (M + M^\top)/2$ and $Q = (M - M^\top)/2$ denote its Hermitian and skew-Hermitian parts, whose off-diagonal blocks are

$$S_{ij} = (K_\rho)_{ij} \frac{C_i + C_j}{2}, \quad Q_{ij} = (K_\rho)_{ij} \frac{C_i - C_j}{2}, \quad (37)$$

and define

$$m_\rho = \min_i \left[d_i^- - \sum_{j \neq i} B_{ij}^S \right], \quad (38)$$

$$M_\rho = \max_i \left[d_i^+ + \sum_{j \neq i} B_{ij}^S \right], \quad (39)$$

$$\eta_\rho = \max_i \sum_{j \neq i} B_{ij}^Q. \quad (40)$$

Theorem 6.5 (Commutator-aware rectangle). *The spectrum satisfies $\text{spec}(C_H C_G) \subset [m_\rho, M_\rho] + i[-\eta_\rho, \eta_\rho]$, and therefore*

$$\varrho(T_{\rho,\alpha}) \leq U_{\text{rect}} := \max_{x \in \{m_\rho, M_\rho\}} \Phi(x), \quad (41)$$

$$\Phi(x) = \sqrt{\left(1 - \frac{\alpha}{2} + \frac{\alpha x}{2}\right)^2 + \left(\frac{\alpha \eta_\rho}{2}\right)^2}.$$

For fixed ρ , the minimizing α is an endpoint of the admissible interval, a stationary point of either quadratic under the square root, or the intersection $4/(2 - m_\rho - M_\rho)$ when that value is admissible.

Proof sketch Real parts of eigenvalues lie in the Rayleigh interval of S , and imaginary parts are

bounded by $\|Q\|_2$. Hermitian block-Gershgorin estimates bound S , while the symmetric matrix of block norms bounds $\|Q\|_2$ by its maximum row sum. Note that $\eta_\rho = 0$ for constant data blocks, in which case the rectangle degenerates to a real interval. $\square$

The row maxima in (38)–(40) can be replaced by a strictly stronger global comparison construction.

Theorem 6.6 (Perron comparison rectangle). *Theorem 6.5 remains valid with*

$$\begin{aligned} m_\rho &= \lambda_{\min}(\text{diag}(d^-) - B^S), \\ M_\rho &= \lambda_{\max}(\text{diag}(d^+) + B^S), \\ \eta_\rho &= \varrho(B^Q). \end{aligned} \quad (42)$$

These bounds are never weaker than their block-row-sum counterparts. They require three symmetric scalar $n \times n$ eigenproblems and never form the $dn \times dn$ iteration matrix. We write $U_{\text{cmp}}(\rho, \alpha)$ for the value of (41) obtained from (42).

Proof sketch For a block vector x , set $y_i = \|x_i\|_2$. Bounding each Hermitian quadratic cross term by $B_{ij}^S y_i y_j$ gives the two scalar Rayleigh quotients. The same block-norm comparison gives $\|Qx\|_2 \leq \|B^Q y\|_2$, hence $\|Q\|_2 \leq \|B^Q\|_2 = \varrho(B^Q)$. Standard bounds of eigenvalues by maximum row sums prove dominance over the earlier rectangle. $\square$

Corollary 6.7 (Sparse truncated-kernel certificate). *Assume periodic boundaries, so that K_ρ is the convolution operator on the n -point grid associated with a scalar kernel k_ρ , indexed by discrete spatial offsets s . Retain offsets in a finite stencil $\mathcal{R}$, write $K_{\rho,\mathcal{R}}$ for the truncated convolution that keeps only those offsets, define the omitted tail $\tau_{\mathcal{R}} = \sum_{s \notin \mathcal{R}} |k_\rho(s)|$, and construct the comparison rectangle of Theorem 6.6 from the retained offsets only, giving $U_{\mathcal{R}}$. Then*

$$\varrho(T_{\rho,\alpha}) \leq U_{\mathcal{R}} + \frac{\alpha}{2} \max_i \|C_i\|_2 \tau_{\mathcal{R}}. \quad (43)$$

For a fixed-radius stencil the comparison matrices are sparse, FFT evaluation of k_ρ costs $O(n \log n)$, and the extremal comparison eigenvalues can be estimated matrix-free.

Algorithm 1 Search-free predict-then-certify

1. Compute the exact rank-one blocks H_i and their distinct curvatures, and combine them with the exact symbol endpoints of G_0 .
 2. Enumerate the joint constant-model algebraic candidates and select (ρ_p, α_p) from the four-corner predictor (Theorems 5.3 and 7.1).
 3. At this single pair, compute C_i , K_{ρ_p} , B^S , B^Q , and the Perron comparison rectangle of Theorem 6.6.
 4. Return $(\rho_p, \alpha_p, U_{\text{cmp}})$ with status **CERTIFIED_USEFUL** if $U_{\text{cmp}} < 1$ and no boundary collapse occurs, and **CERTIFIED_CONSERVATIVE** or **UNCERTIFIABLE** otherwise.
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Proof sketch Young’s convolution inequality gives $\|K_\rho - K_{\rho, \mathcal{R}}\|_2 \leq \tau_{\mathcal{R}}$; multiplication by C_H and the affine oADMM scaling give the displayed correction. $\square$

This certificate is rigorous for every truncation radius, including radius zero; tightness rather than validity determines whether it is actionable.

6.3 Predict, then certify

The preceding bounds motivate a two-stage method, summarized in Algorithm 1. First, extract h_+ and the two regularizer-symbol endpoints and use the four-corner candidates to predict (ρ_p, α_p) . Second, evaluate the comparison rectangle once at that pair.

Corollary 6.8 (Predict-then-certify). *The prediction stage uses no full spectral optimization. If the comparison bound satisfies $U_{\text{cmp}}(\rho_p, \alpha_p) < 1$, then the full variable-coefficient oADMM iteration converges linearly, irrespective of whether the pixel-curvature predictor is itself an enclosure.*

This separation matters: a conservative certificate can validate a good predictor without being allowed to drag the minimizer toward a poor boundary solution.

6.4 Diagnosing the nonnormal regime

Finally, the resolvent identity gives the explicit commutator estimate

$$\|[C_H, C_G]\|_2 \leq \frac{4\rho^2 \| [H, G] \|_2}{(\lambda_{\min}(H) + \rho)^2 (\lambda_{\min}(G) + \rho)^2}, \quad (44)$$

together with $[H, G]_{ij} = (G_0)_{ij}(H_i - H_j)$, where the left-hand side is the (i, j) velocity block of the commutator. Nonnormality is therefore controlled by local rather than global Hessian differences.

For the rank-one model, a block rotation $Q_i \in \mathbb{R}^{d \times d}$ aligned with q_i diagonalizes H_i exactly. In the rotated coordinates the regularizer block is $(G_0)_{ij} Q_i^\top Q_j$, whose departure from the identity is $2|\sin((\phi_i - \phi_j)/2)|$, where ϕ_i is the gradient orientation at pixel i . The implementation therefore records neighborwise magnitude-squared variation together with the unoriented-angle statistic $|\sin((\phi_i - \phi_j)/2)|$; the sign of q_i is immaterial because only $q_i q_i^\top$ enters. These statistics diagnose the commutator regime without forming a global mean Hessian.

7 Finite Four-Corner Parameter Selection

We use *search-free* to mean that no grid search, no gradient descent in (ρ, α) , no repeated ADMM trial, and no repeated full iteration-matrix eigen-solve is performed. The practical method extracts curvature and symbol endpoints once, generates a finite algebraic candidate set, and evaluates the constant-model predictor envelope; any certificate is evaluated separately at the selected pair.

7.1 Globally joint constant-gradient selector

For each corner $c = (h_c, g_c) \in \mathcal{C}$, write $\theta_c(\rho) := \theta(h_c, g_c; \rho)$. For fixed ρ , let

$$\theta_-(\rho) = \min_{c \in \mathcal{C}} \theta_c(\rho), \quad \theta_+(\rho) = \max_{c \in \mathcal{C}} \theta_c(\rho), \quad (45)$$

On $0 < \alpha \leq 2$, exact minimization of (29) gives

$$\alpha_*(\rho) = \min \left\{ 2, \frac{2}{2 - \theta_- - \theta_+} \right\}. \quad (46)$$

After eliminating α , the reduced objective equals $(\theta_+ - \theta_-)/(2 - \theta_- - \theta_+)$ when $\theta_- + \theta_+ \leq 1$, and $2\theta_+ - 1$ when $\theta_- + \theta_+ \geq 1$.

Theorem 7.1 (Exact search-free constant-gradient tuning). *On a compact positive penalty interval, a globally optimal pair (ρ, α) is obtained by evaluating a finite algebraic set comprising the interval endpoints, the pairwise corner-branch intersections, the roots of $\theta_- + \theta_+ = 1$, and the stationary roots of the two rational objectives above for every active-corner pair. After clearing positive quadratic denominators, branch intersections have degree at most three, transitions degree at most four, and stationary equations degree at most six.*

Proof sketch Between consecutive branch intersections the active minimum and maximum are fixed rational functions, so a minimum of the continuous piecewise-rational reduced objective is an endpoint, a transition, or a stationary root. Enumerating all corner pairs is a finite superset and removes the need to determine activity symbolically. The implementation verifies companion-matrix roots by polynomial residual and interval membership. $\square$

7.2 Standard ADMM penalty

For $\alpha = 1$ all local modal factors are nonnegative, so the certified envelope of Theorem 6.2 reduces to

$$\mathcal{U}_1(\rho) = \max_{c \in \mathcal{C}} \theta_c(\rho) + \frac{\delta}{\rho}, \quad (47)$$

where $\delta = \delta_H + \delta_G$ is the patch-surrogate discrepancy from (30); the deployed four-corner predictor uses the special case $\delta = 0$. Select a physically meaningful compact interval $\mathcal{I} = [\rho_-, \rho_+]$, for example

$$\rho_- = \epsilon_{\text{num}}, \quad \rho_+ = \max_{c \in \mathcal{C}} \max(h_c, g_c) + \delta, \quad (48)$$

with a small positive numerical floor ϵ_{num} .

Two corner branches $c_1 = (h_1, g_1)$ and $c_2 = (h_2, g_2)$ intersect at explicitly computable points: clearing denominators in $\theta(h_1, g_1; \rho) = \theta(h_2, g_2; \rho)$ gives

$$\rho(A_{12}\rho^2 + B_{12}) = 0, \quad (49)$$

where

$$\begin{aligned} A_{12} &= (h_2 + g_2) - (h_1 + g_1), \\ B_{12} &= g_1 g_2 (h_1 - h_2) + h_1 h_2 (g_1 - g_2), \end{aligned} \quad (50)$$

so that a positive intersection is $\rho_{12} = \sqrt{-B_{12}/A_{12}}$ whenever the radicand is positive. On an interval where corner $c = (h, g)$ is active, the stationary points of the corrected branch solve the quartic

$$(h + g)\rho^2(\rho^2 - hg) - \delta(\rho + h)^2(\rho + g)^2 = 0. \quad (51)$$

Theorem 7.2 (Finite-candidate ADMM selector). *Let $\mathcal{S}_\rho$ contain the interval endpoints ρ_- and ρ_+ , all positive pairwise intersection roots of (49) in $\mathcal{I}$, and all positive roots of (51) in $\mathcal{I}$ that lie on a subinterval where the associated corner is active. Then every global minimizer of $\mathcal{U}_1$ on $\mathcal{I}$ belongs to $\mathcal{S}_\rho$, so that*

$$\rho_{\text{SF}} = \arg \min_{\rho \in \mathcal{S}_\rho} \mathcal{U}_1(\rho) \quad (52)$$

minimizes the certified standard-ADMM envelope exactly, without iterative parameter search.

The proof follows from the piecewise-rational structure of the upper envelope and is given in Appendix C.

7.3 Conditional search-free relaxation

After selecting ρ_{SF} , abbreviate $\theta_- = \theta_-(\rho_{\text{SF}})$ and $\theta_+ = \theta_+(\rho_{\text{SF}})$, with $0 \leq \theta_- \leq \theta_+ < 1$ for a well-posed local model. The certified oADMM envelope conditional on ρ_{SF} is

$$\begin{aligned} F(\alpha) &= \max \{ |1 - \alpha + \alpha\theta_-|, |1 - \alpha + \alpha\theta_+| \} \\ &\quad + \frac{\alpha\delta}{\rho_{\text{SF}}}, \end{aligned} \quad (53)$$

which is convex and piecewise linear in α .

Theorem 7.3 (Finite-candidate oADMM relaxation). *Let the admissible interval be $[\alpha_-, \alpha_+] \subset (0, 2]$. A minimizer of (53) belongs to*

$$\begin{aligned} \mathcal{S}_\alpha &= \left\{ \alpha_-, \alpha_+, \frac{1}{1-\theta_-}, \frac{1}{1-\theta_+}, \frac{2}{2-\theta_--\theta_+} \right\} \\ &\quad \cap [\alpha_-, \alpha_+], \end{aligned} \quad (54)$$

Algorithm 2 Certified-envelope minimization (optional baseline)

1. Compute image gradients and the exact block data Hessian H .
 2. Partition the grid; construct H_r and a transform-diagonalizable G_r .
 3. Extract $h_{r,j}$ and the regularizer endpoints g_r^-, g_r^+ ; form $\mathcal{C}$ as in (28).
 4. Compute or bound δ_H and δ_G .
 5. Generate $\mathcal{S}_\rho$ from the interval endpoints, the corner intersections, and the active quartic roots.
 6. Evaluate $\mathcal{U}_1$ and select ρ_{SF} by (52).
 7. Form $\mathcal{S}_\alpha$ and select α_{SF} by (55).
 8. Run oADMM once with the selected parameters, optionally reporting $\mathcal{U}(\rho_{\text{SF}}, \alpha_{\text{SF}})$ as a certificate.
-

so that

$$\alpha_{\text{SF}} = \arg \min_{\alpha \in \mathcal{S}_\alpha} F(\alpha) \quad (55)$$

conditionally minimizes the certified oADMM envelope at ρ_{SF} .

When $\delta = 0$ and the unconstrained equioscillation point lies in $(0, 2]$, (54) recovers the familiar formula $\alpha_0 = 2/(2 - \theta_- - \theta_+)$, which then coincides with $\alpha_*(\rho_{\text{SF}})$ from (46). A nonzero perturbation term may instead favor a smaller relaxation.

7.4 Algorithm summary

Algorithm 2 collects the certified-envelope variant of the selector, which minimizes $\mathcal{U}_1$ with $\delta > 0$ and returns $(\rho_{\text{SF}}, \alpha_{\text{SF}})$; it is retained as an optional baseline. The deployed predictor used in all timing experiments is the $\delta = 0$ special case of the joint four-corner selector of Theorems 5.3 and 7.1, whose selected pair is denoted (ρ_p, α_p) in Algorithm 1 and in the figures.

8 Implementation and Experimental Protocol

8.1 Patch construction

Candidate patch strategies include fixed cubic patches, multiresolution patches matched to the registration pyramid, and adaptive patches that are split wherever gradient magnitude or orientation

varies beyond a threshold. A robust representative $\bar{q}_r$ can be obtained from the spatial median of gradients. Because H is block diagonal, the direct block surrogate satisfies

$$\delta_H = \max_i \|\mu q_i q_i^\top - H_{r(i)}\|_2, \quad (56)$$

where $r(i)$ is the patch containing pixel i .

8.2 Regularizer symbols

For periodic or cosine-transform boundary closures the endpoints $g_r^\pm$ are known analytically. For an order- p Sobolev regularizer with symbol

$$g(\xi) = \left(\gamma + \frac{4\beta}{(\Delta x)^2} \sum_{\ell=1}^d \sin^2(\xi_\ell/2) \right)^p, \quad (57)$$

the endpoints are

$$g_- = \gamma^p, \quad g_+ = \left(\gamma + \frac{4\beta d}{(\Delta x)^2} \right)^p. \quad (58)$$

The patch-boundary discrepancy δ_G can be reduced by using overlapping patches or cosine closures rather than disconnected periodic patches.

8.3 Computational complexity

For the deployed global predictor, a single image scan obtains h_+ in $O(n)$ time. The four branches then yield a constant-size algebraic candidate calculation, whose cost is independent of the image size. No full iteration matrix is formed, and no numerical optimization over ρ or repeated ADMM trial is performed. Patchwise constructions are retained only for the optional rate analysis of Section 6.

8.4 End-to-end two-dimensional solver

The reference implementation uses a three-level Gaussian pyramid, spatial image gradients, Gauss–Newton optical-flow linearization, Sherman–Morrison solves of the rank-one data blocks, and FFT solves of the scalar Sobolev regularizer. Velocity increments are composed with the current displacement. A backtracking line search requires objective decrease and a discrete Jacobian above 0.05. Because interpolation between pyramid levels can itself introduce grid-scale folds, the upsampled

field is audited before the next update; if necessary, only its non-translational component is halved until the same Jacobian floor holds. The one-shot policy estimates parameters from the initial gradient on the finest experimental grid and reuses them at every outer iteration and every pyramid level. All baselines share the same initialization, objective, stopping tolerances, line search, pyramid, and topology safeguards.

9 Results

We first validate the spectral reduction on problems for which the full iteration spectrum is available, then assess prediction quality for smoothly varying coefficients, and finally evaluate end-to-end registration performance. All methods within a comparison use identical solver settings, initialization, stopping rules, and topology safeguards. Complete software, hardware, seed, and command details are provided in the supplementary material. Small-system oracle values use direct eigendecomposition and multistart scalar optimization; this numerical optimization is diagnostic only and is never used for tuning.

Two relative quality measures recur below. Writing ϱ_{act} for the exact spectral radius attained at the selected parameters, ϱ_{ora} for the radius attained at oracle-optimal parameters, and U for the particular rigorous bound under discussion in that experiment (typically U_{cmp} , or $\mathcal{U}$ for the additive patch certificate), the *parameter gap* is $(\varrho_{\text{act}} - \varrho_{\text{ora}})/\varrho_{\text{ora}}$ and the *enclosure gap* is $(U - \varrho_{\text{act}})/\varrho_{\text{act}}$. Both are dimensionless; smaller is better.

9.1 Exactness of the four-corner reduction

We used $n \times n$ periodic grids, data curvatures $(h_-, h_+) = (0.2, 2.3)$ corresponding to $(\nu, \mu \max_i \|q_i\|^2) = (0.2, 2.1)$ in (26), and first-order regularization with $(\beta, \gamma) = (0.2, 0.1)$. The finite selector returned $(\rho, \alpha) = (0.543512, 1.898152)$ on every grid, and Table 1 shows that the local envelope agrees with the directly computed spectral radius to machine precision, as predicted by Theorems 5.1 and 5.3. Panel (a) of Fig. 1 displays the same agreement.

Table 1 Constant-coefficient verification. The local envelope agrees with the directly computed spectral radius to machine precision on every grid.

n	Direct radius $\varrho(T)$	$ U_{\text{LFA}} - \varrho(T) $
4	0.35313044	$< 10^{-15}$
6	0.35313044	$< 10^{-15}$
8	0.35313044	$< 10^{-15}$

Table 2 Controlled structured-envelope results over 18 cases. Gaps are the medians of the relative enclosure and parameter gaps defined above; a collapse is a selected pair at the boundary of the admissible set.

Method	Median gap		
	Enclosure	Parameter	Collapses
Block resolvent	0.166	0.926	6
Block Gershgorin	0.644	0.560	0
Commutator rectangle	0.602	0.431	0
Comparison rectangle	0.595	0.386	0

9.2 Comparison of signed enclosures

Minimizing the global additive perturbation bound of Theorem 6.2 produced over-conservative relaxation parameters in the controlled variable-coefficient tests, and it collapsed to the boundary of the admissible set in six of eighteen cases. The signed enclosures of Theorems 6.4, 6.5, and 6.6 retain the spectral sign information that the additive bound discards. Every enclosure contained the exact radius. The Perron rectangle of Theorem 6.6 was the tightest, with median enclosure and parameter gaps of 0.595 and 0.386 (Table 2), but it remains too loose to serve as an online tuning objective. Detailed constructions and stress tests are reported in the supplementary material.

9.3 Prediction quality under smooth coefficient variation

We next separated periodic magnitude variation, smooth orientation variation, a sharp orientation interface, and Gaussian-smoothed image gradients on 3×3 grids, giving ten cases of which nine are smooth. The four-corner predictor achieved a median parameter gap of 0.057 overall and 0.027 on the smooth cases, with no collapse; see panel (b) of Fig. 1. At the selected parameters, the rigorous comparison bound was below one in all ten cases,

so Corollary 6.8 applies throughout. Its median enclosure gap was 1.29, confirming validity but also the substantial conservatism that restricts these bounds to diagnostic rate analysis (panel (c)).

9.4 Scaling of the truncated certificate

The rigorous truncated-kernel certificate of Corollary 6.7 produced no violation in 20 tests spanning 8^2 to 64^2 grids and five truncation radii. Actual matrix-free radii were approximately 0.269, whereas the bounds ranged from 0.638 to 0.681, corresponding to enclosure gaps of 1.38 to 1.54. The most expensive 64^2 retained-radius case took 81.4 seconds on CPU, and a relaxed 128^2 Arnoldi certificate run did not complete within the allocated diagnostic window. The construction therefore establishes a scalable sparse representation in principle, but the present eigensolver and bound are not a practical registration-time certificate. We retain it as a small-problem diagnostic and use the predictor alone in the end-to-end timing claims.

9.5 Controlled real-image pilot

We next tested the multiresolution solver on a public-domain retinal image and an unrestricted immunohistochemistry image, each subjected to three held-out smooth deformations. Deformation cases 1–2 were used to select a single global manual baseline $(\rho, \alpha) = (0.1, 1)$; cases 3–5 were not used for tuning. This is a controlled real-content pilot rather than a landmark benchmark. Against that validation-selected baseline, a single predictor evaluation from the initial full-resolution gradient reduced the median inner-iteration count by 44.4%. Across five repeated, order-alternated timings (30 paired runs), the median total-time reduction was 41.7% (bootstrap 95% interval $[40.9, 42.6]\%$), tuning consumed 1.58% of proposed-method runtime, and the median relative change in mean squared error was -0.0068% . No tested case had a nonpositive discrete Jacobian. The experiment therefore passes every predeclared pilot gate (Table 3), but its two images and controlled warps are insufficient for a clinical or benchmark claim.

Table 3 Controlled real-image pilot against the predeclared acceptance gates; medians over paired runs. The accuracy gate admits at most a 1% degradation in mean squared error.

Metric	Observed	Gate
Inner-iteration reduction	44.4%	$\geq 15\%$
Total-time reduction	41.7%	$\geq 10\%$
Relative MSE change	-0.0068%	$\leq 1\%$
Selection overhead	1.58%	$< 2\%$
Nonpositive Jacobians	0/6	0

9.6 CIMA histology landmark benchmark

We then evaluated the public lung-lesion-3 sample distributed with the CIMA histology-landmark repository [3], comprising five differently stained sections and all ten unordered image pairs. Images supplied at 5% scale were converted to a common structural channel by grayscale conversion, CLAHE, Sobel magnitude, Gaussian smoothing, and fixed percentile normalization, and were then resized to 256^2 . Manual landmarks provided at 50% scale were rescaled geometrically and used only for evaluation. Every method used the same image-only robust translation initialization, selected from phase, centroid, and clustered SIFT proposals on a 128^2 image. A pyramid-transition safeguard damped only the non-translational displacement when interpolation threatened the discrete Jacobian floor.

The proposed practical policy evaluated the four-corner selector once from the initial full-resolution gradient and reused its parameters through all levels. The strongest fixed baseline, $(\rho, \alpha) = (0.1, 1)$, had been selected on the separate controlled real-content validation cases and was frozen before CIMA. Across five order-alternated repetitions of all ten pairs (50 paired runs), the predictor reduced median total runtime by 20.1% (bootstrap 95% interval $19.4\text{--}20.7\%$), solver time by 23.5%, and iterations by 30.1%; selection used 1.16% of proposed-method runtime. Median target registration error (TRE) was 7.33 pixels, or 2.02% of the image diagonal, and differed by less than 3×10^{-5} pixels between the predictor and the fixed baseline (Table 4). The median landmark-error improvement relative to the unregistered pairs was 36.5%, and no global or interior nonpositive Jacobian was observed in the final protocol. One

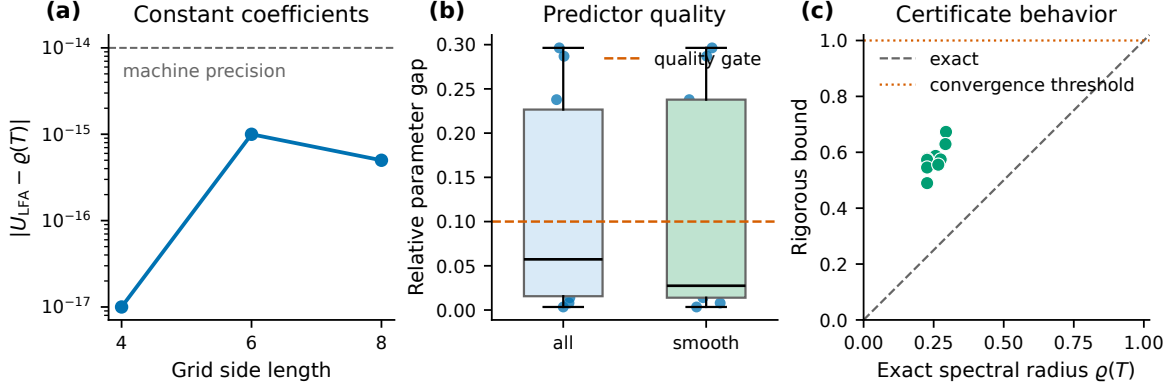


Fig. 1 Spectral validation. (a) The Fourier envelope is exact in constant-coefficient periodic systems, with the deviation from the directly computed radius at the level of machine precision. (b) The four-corner predictor has a small relative parameter gap in smooth variable-coefficient cases; the dashed line is the predeclared quality gate of 0.10. (c) The signed rate certificate is rigorous, since every point lies above the identity line, but conservative.

Table 4 CIMA landmark results. Accuracy medians are taken over the ten pairs and timing medians over 50 paired runs; the proposed method is in bold. “BB proxy” is the safeguarded Barzilai–Borwein rule and “external fixed” is the pair selected on the separate validation cases of Table 3.

Method	Iterations	Time (s)	TRE (px)
Fixed ADMM (1, 1)	1910.5	12.09	7.3310
Fixed oADMM (1, 1.8)	1129.5	7.37	7.3308
Residual balancing	1124.5	7.40	7.3310
BB proxy	435.5	3.44	7.3306
External fixed (0.1, 1)	303.5	2.53	7.3306
Four-corner	210.0	2.02	7.3306

difficult stain pair retained a TRE above 22 pixels, identifying global initialization and cross-stain representation, rather than ADMM convergence, as the principal failure mode. Phase-only and unprotected pyramid runs on the same sample exposed the initialization and upsampling failures that motivated the final shared safeguards; those raw runs are preserved in the repository. The final CIMA estimate is therefore a reproducible exploratory benchmark rather than a held-out confirmatory result.

For a diagnostic comparison in the style of Song et al. [12], we formed the exact variable-coefficient reduced operator on an 8×8 surrogate for every pair, optimized ρ numerically with the exact conditional α , and then reused the result in the full solve. This required a median of 97 spectral objective evaluations and 2.18 seconds of tuning. Its median full-solve iteration count was 251 versus 210 for the four-corner predictor, and its total runtime was

4.39 versus 2.02 seconds, with indistinguishable TRE. Because the 8×8 spectral model is a coarse surrogate, this comparison illustrates overhead and model-transfer behavior rather than a claim of superiority to full-resolution spectral optimization.

9.7 FIRE retinal landmark benchmark

We evaluated all 134 landmarked pairs in the public FIRE fundus registration benchmark [8], comprising 14 category-A, 49 category-P, and 71 category-S pairs. The released archive was integrity-tested with 7z; images were converted to a foreground-percentile-normalized green channel and resized to 256^2 . Supplied control points were scaled geometrically and used only for evaluation. All methods used identical phase-correlation initialization, three-level pyramid, objective, stopping criteria, and a single CPU BLAS/OpenMP thread. The four-corner predictor was evaluated once from the initial 256^2 gradient and reused throughout the solve. We compare against the fixed external pair $(\rho, \alpha) = (0.1, 1)$ without claiming that it is a FIRE-tuned best baseline.

Across all pairs, median predictor iterations were 245 versus 412 for the fixed pair, a paired median reduction of 39.6%. Median total time was 4.774 versus 6.680 seconds, a 34.4% paired reduction (bootstrap 95% interval 30.7–35.7%), and tuning accounted for 1.2% of proposed total time. The safeguarded BB proxy required a median of 623.5 iterations and 9.435 seconds; the predictor

was faster on 94.0% of pairs, with a median paired reduction of 51.7%. Median TRE differed from the fixed pair by -1.4×10^{-5} pixels, with a maximum absolute difference of 3.9×10^{-4} pixels, and no method produced a nonpositive Jacobian. Table 5 summarizes these results and Fig. 2 resolves them per pair and per benchmark category.

Reported percentage reductions are medians of pairwise percentage changes and therefore need not equal the percentage differences between the separately reported method medians of Table 5.

9.8 One-shot reuse ablation

On a balanced 18-pair FIRE ablation, retuning at every pyramid level reduced the median inner-iteration count from 243 to 222, but increased median total time from 2.651 to 3.895 seconds because of the tuning overhead. One-shot reuse is therefore the recommended deployment policy, and it is the policy used in all results reported above.

10 Discussion

10.1 Convergence, prediction, and certification

For a well-posed quadratic subproblem, Theorem 4.2 already establishes convergence for every admissible pair, so tuning is a matter of speed rather than of stability. The four-corner predictor selects a pair expected to have a small contraction factor; it is not claimed to enclose the exact variable-coefficient spectrum. The signed enclosures of Section 6 are optional upper bounds on that contraction factor and are never evaluated by the timed registration algorithm.

10.2 Relation to adaptive and spectral tuning

Song et al. [12] optimize the penalty numerically for general linear quadratic problems and derive a conditional relaxation formula. Here, the registration-specific rank-one geometry reduces the constant-model problem to a finite joint selector without any repeated full-spectrum optimization. The CIMA 8×8 surrogate experiment concerns overhead and model transfer only; it is not a comparison with full-resolution spectral optimization.

10.3 Limitations

Outside the constant-gradient setting, the selected parameters are not claimed to minimize the exact full spectral radius, and the finite penalty candidates used with the signed enclosures are inherited from the local model rather than proved to minimize those enclosures globally. Most importantly, the empirically successful four-corner rule is predictive rather than rigorous when gradient orientations vary.

The certificates themselves remain loose: even the Perron comparison bound discards directional cancellation in the dense C_G because it works with block norms. Most exact spectral experiments are consequently small. The CIMA validation uses a single five-image tissue set at reduced resolution, and the FIRE study uses a 256^2 CPU protocol against an externally validation-selected fixed comparator, residual balancing, and a safeguarded Barzilai–Borwein proxy; that proxy is not a complete reproduction of every published adaptive ADMM variant, and the study does not establish full-resolution clinical throughput. Finally, the theory concerns a convex linearized subproblem and not the global convergence of the outer diffeomorphic scheme.

11 Conclusion

For well-posed linearized registration subproblems, all admissible ADMM parameters are already contractive, so the practical challenge is to select a rapidly convergent pair. We showed that rank-one data-Hessian structure reduces the constant-model tuning problem exactly to four curvature–symbol corners, which enables a finite algebraic one-shot predictor that requires no numerical spectral search. On all 134 FIRE pairs at 256^2 , the predictor reduced inner iterations by 39.6% and median pairwise runtime by 34.4% at essentially unchanged landmark accuracy and with no nonpositive discrete Jacobians, and it reduced matched-protocol runtime by 20.1% on CIMA. Signed variable-coefficient enclosures provide rigorous but currently conservative optional rate bounds that the deployed algorithm does not require. Extending the predictor to three-dimensional volumetric registration, and tightening the enclosures so that they become usable as online tuning objectives, are the natural directions for future work.

Table 5 FIRE benchmark, 134 pairs at 256^2 , under identical initialization and solver protocol. Medians are reported; the proposed method is in bold. All median TRE values agree to within 4×10^{-4} pixels, so the comparison is one of cost at matched accuracy. Naming follows Table 4.

Method	Iterations	Total time (s)	TRE (px)	TRE/diagonal (%)
Fixed oADMM (1.0, 1.8)	1529.5	23.471	5.50194	1.51971
Residual balancing	1434.0	22.325	5.50211	1.51976
BB proxy	623.5	9.435	5.50175	1.51966
External fixed (0.1, 1.0)	412.0	6.680	5.50178	1.51967
Four-corner, one-shot	245.0	4.774	5.50177	1.51966

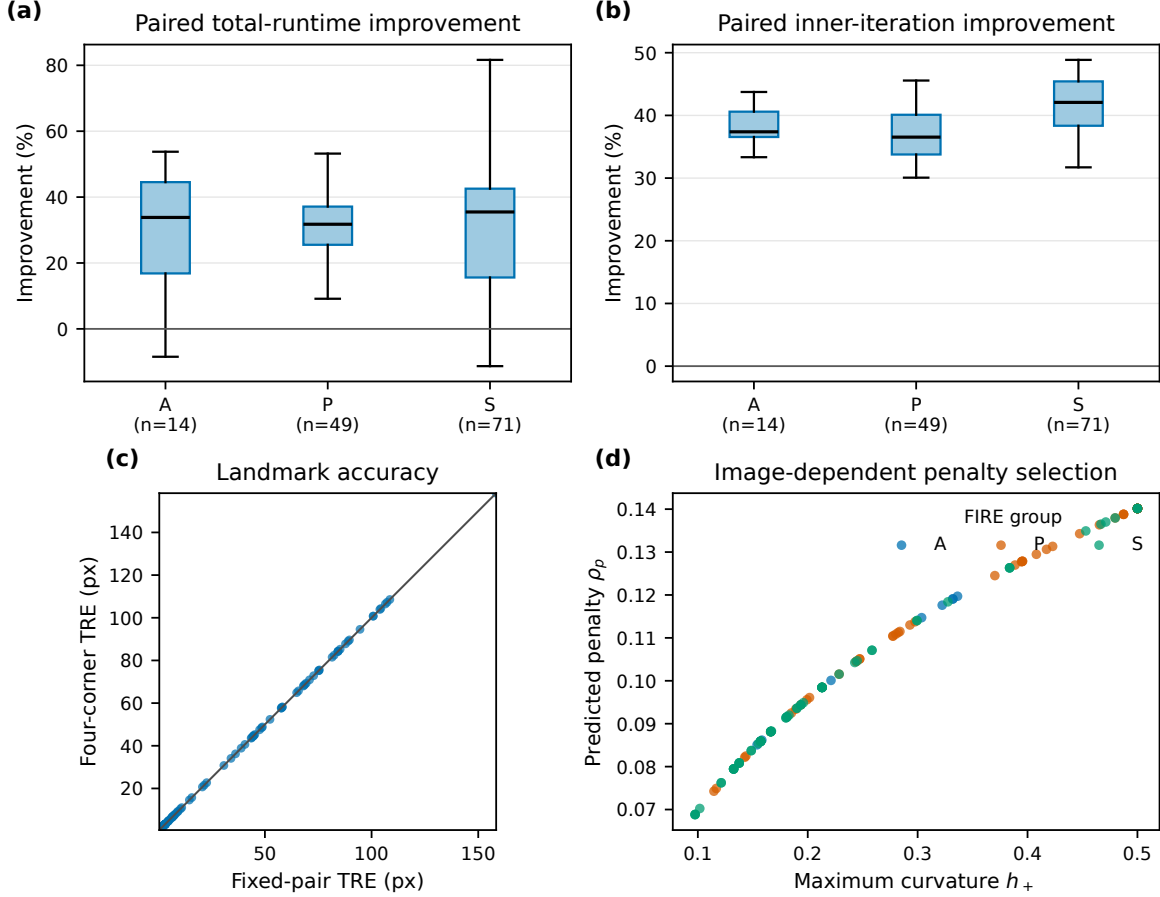


Fig. 2 FIRE benchmark at 256×256 ($n = 134$). (a) Pairwise total-runtime reduction of one-shot four-corner tuning relative to the externally validation-selected fixed pair, stratified by the A, P, and S categories. (b) Corresponding inner-iteration reduction. (c) Landmark TRE for the two methods; the identity line indicates unchanged accuracy. (d) The predicted penalty ρ_p varies with the maximum data curvature h_+ , demonstrating image-dependent tuning. Percentage reductions are computed per pair before the category summaries.

Supplementary information. Supplementary Information, containing the detailed enclosure constructions, stress tests, and full software, hardware, seed, and command details, is supplied as Online Resource 1.

Declarations

- **Funding.** No specific funding was received for this work.

- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable; the study uses only publicly available, de-identified image datasets.
- **Consent for publication.** Not applicable.
- **Data availability.** The FIRE [8] and CIMA [3] datasets are publicly available from their original providers. The FIRE archive is not redistributed here.
- **Materials availability.** Not applicable.
- **Code availability.** Source code, tests, configurations, the raw processed JSON and CSV outputs, and the FIRE and CIMA preparation and evaluation scripts are available at <https://github.com/xhxuciedu/admm-registration>.
- **Author contribution.** Katherine Xie and Jie Wu conducted the research and wrote and edited the manuscript. Xiaohui Xie conceived the work and edited the manuscript.

Appendix A Proof of the Reduced Cayley Representation

For standard ADMM, the homogeneous updates are

$$v^{k+1} = P_H(w^k - u^k), \quad w^{k+1} = P_G(v^{k+1} + u^k). \quad (\text{A1})$$

Eliminating v^{k+1} gives the state iteration

$$\begin{bmatrix} w^{k+1} \\ u^{k+1} \end{bmatrix} = T_{\rho,1} \begin{bmatrix} w^k \\ u^k \end{bmatrix}, \quad (\text{A2})$$

where

$$T_{\rho,1} = \begin{bmatrix} P_G P_H & P_G(I - P_H) \\ (I - P_G)P_H & (I - P_G)(I - P_H) \end{bmatrix}, \quad (\text{A3})$$

which factors as

$$T_{\rho,1} = \begin{bmatrix} P_G \\ I - P_G \end{bmatrix} \begin{bmatrix} P_H & I - P_H \end{bmatrix}. \quad (\text{A4})$$

The nonzero eigenvalues of AB and BA coincide, so the nonzero spectrum of $T_{\rho,1}$ equals that of $P_H P_G + (I - P_H)(I - P_G)$. Substituting $P_H =$

$(I - C_H)/2$ and $P_G = (I - C_G)/2$ yields

$$P_H P_G + (I - P_H)(I - P_G) = \frac{1}{2}(I + C_H C_G), \quad (\text{A5})$$

which is (15). The relaxed formula follows by inserting $\hat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha)w^k$ from (10) and collecting the same reduced state, giving

$$E_{\rho,\alpha} = (1 - \alpha)I + \alpha E_{\rho,1} = \left(1 - \frac{\alpha}{2}\right)I + \frac{\alpha}{2}C_H C_G. \quad (\text{A6})$$

Appendix B Proof of the Perturbation Bound

For $C_A = \text{Id} - 2\rho(A + \rho\text{Id})^{-1}$ we have

$$\begin{aligned} C_A - C_B &= 2\rho[(B + \rho\text{Id})^{-1} - (A + \rho\text{Id})^{-1}] \\ &= 2\rho(A + \rho\text{Id})^{-1}(A - B)(B + \rho\text{Id})^{-1}. \end{aligned} \quad (\text{B7})$$

Since $A, B \succeq 0$ imply $\|(A + \rho\text{Id})^{-1}\|_2 \leq \rho^{-1}$, this proves Lemma 6.1. Next, using $\|C_A\|_2 \leq 1$ for $A \succeq 0$,

$$\begin{aligned} \|C_H C_G - C_{\tilde{H}} C_{\tilde{G}}\|_2 &\leq \|(C_H - C_{\tilde{H}})C_G\|_2 + \|C_{\tilde{H}}(C_G - C_{\tilde{G}})\|_2 \\ &\leq \frac{2}{\rho}(\delta_H + \delta_G). \end{aligned} \quad (\text{B8})$$

Therefore

$$\begin{aligned} \varrho(E_{\rho,\alpha}) &\leq \|E_{\rho,\alpha}\|_2 \\ &\leq \|\tilde{E}_{\rho,\alpha}\|_2 + \frac{\alpha}{2}\|C_H C_G - C_{\tilde{H}} C_{\tilde{G}}\|_2 \\ &\leq U_{\text{LFA}}(\rho, \alpha) + \frac{\alpha}{\rho}(\delta_H + \delta_G), \end{aligned} \quad (\text{B9})$$

which is (32).

Appendix C Proof of the Finite Candidate Sets

Each corner branch θ_c is rational and smooth for $\rho > 0$. The pointwise maximum can change its active branch only where two branches intersect, which gives (49), and between consecutive intersections exactly one branch is active. On such a

subinterval, differentiating $\theta(h, g; \rho) + \delta/\rho$ gives

$$\frac{(h+g)(\rho^2 - hg)}{(\rho+h)^2(\rho+g)^2} - \frac{\delta}{\rho^2}, \quad (\text{C10})$$

and clearing the positive denominators yields (51). A global minimum of a continuous piecewise-smooth function on a compact interval occurs at an interval endpoint, at a nonsmooth branch intersection, or at a stationary point of an active smooth branch, which proves Theorem 7.2.

For Theorem 7.3, note that the image of $\theta \in [\theta_-, \theta_+]$ under $\theta \mapsto 1 - \alpha + \alpha\theta$ is an interval, so the maximum absolute value occurs at one of its endpoints. The resulting maximum of two absolute affine functions, plus a linear function of α , is convex and piecewise linear. Its slope changes only at the two zero crossings, at the equioscillation intersection, and at the admissible interval endpoints, which proves (54).

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